

Analyzing U.S. County-Level Health Outcomes

The Role of Lifestyle, Environment, and Access to Care

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Introduction

Why do some counties have far worse health outcomes than others?

Motivation

- Premature death rates vary by 4x across U.S. counties
- Lifestyle factors like inactivity and smoking are modifiable
- Access to care and insurance coverage differ dramatically by region
- Understanding these patterns can inform public health policy

Research Questions

- Which behavioral and environmental factors most strongly correlate with premature death rates?
- Do counties cluster into distinct health risk profiles?
- Which states have the best and worst health outcomes?

Data Sources

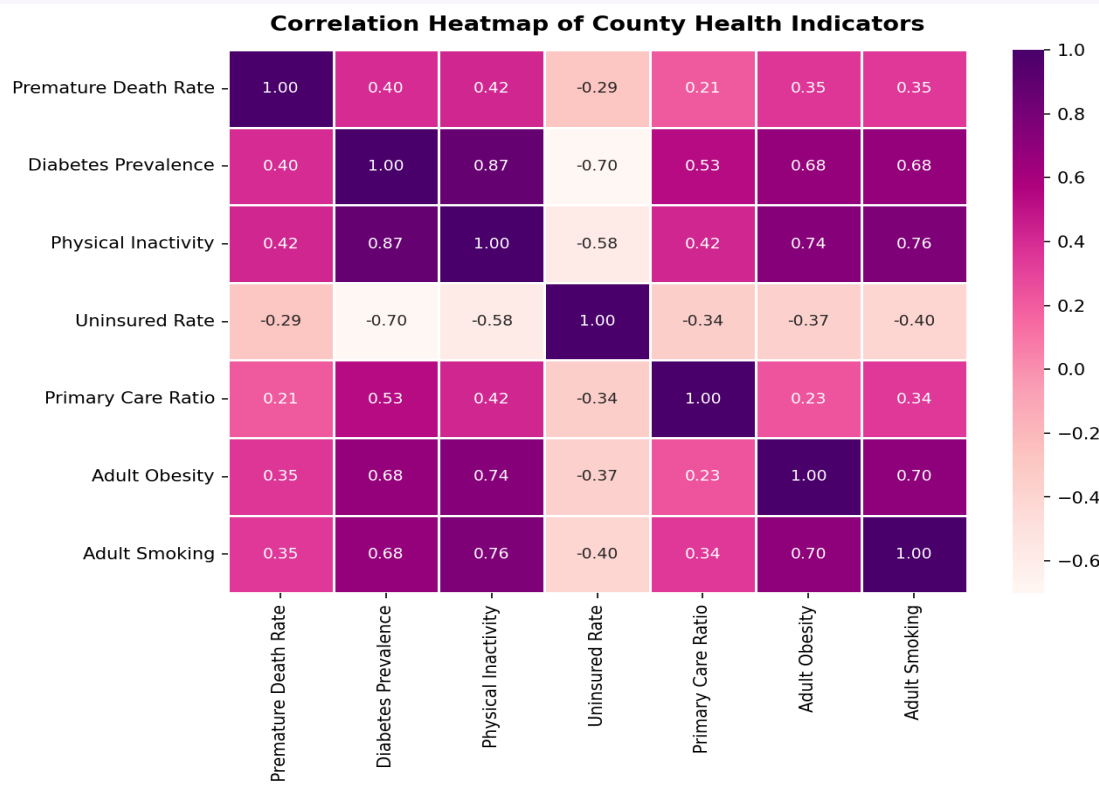
Three datasets merged on county FIPS codes

#	Dataset	Source / Type	Key Fields	Size
1	County Health Rankings 2025	countyhealthrankings.org / CSV File	Premature death, diabetes, physical inactivity, uninsured rate, obesity, smoking	~3,200 counties
2	CDC PLACES API	data.cdc.gov / API (JSON)	Chronic disease measures by county — obesity, smoking, checkup rates	500+ measures, 3,100+ counties
3	EPA Air Quality Annual Summary	aqs.epa.gov / CSV File	Median AQI, days good/unhealthy, max AQI per county	~1,000+ counties

All datasets joined on 5-digit county FIPS codes. Final analysis dataset: 2,843 counties after removing missing values.

Results: Correlation Analysis

Which factors most strongly predict premature death rates?



Correlation with Premature Death Rate

Physical Inactivity $r = 0.42$

Diabetes Prevalence $r = 0.40$

Adult Obesity $r = 0.35$

Adult Smoking $r = 0.35$

Uninsured Rate $r = 0.29$

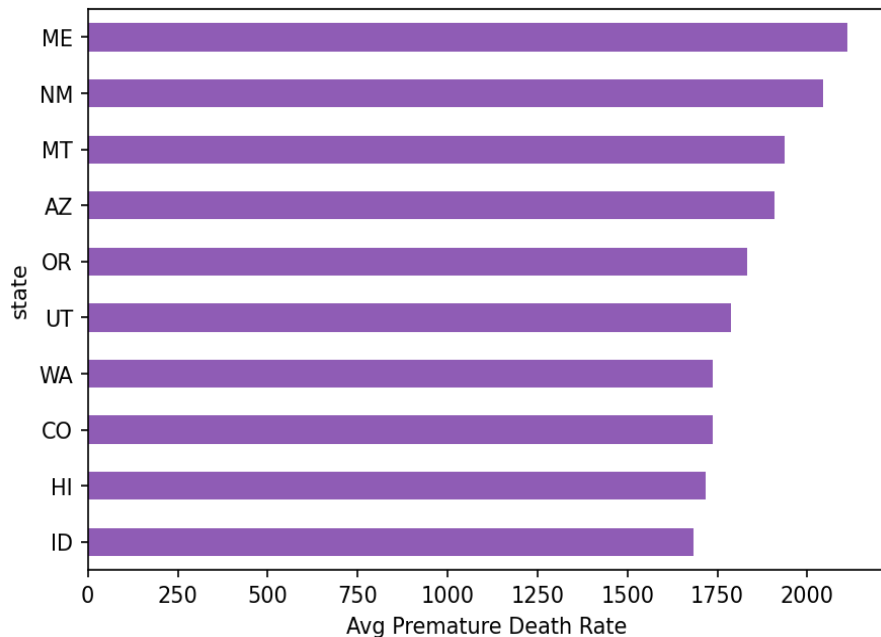
All correlations statistically significant ($p < 0.001$)

Results: State-Level Patterns

Healthiest vs. least healthy states by average premature death rate

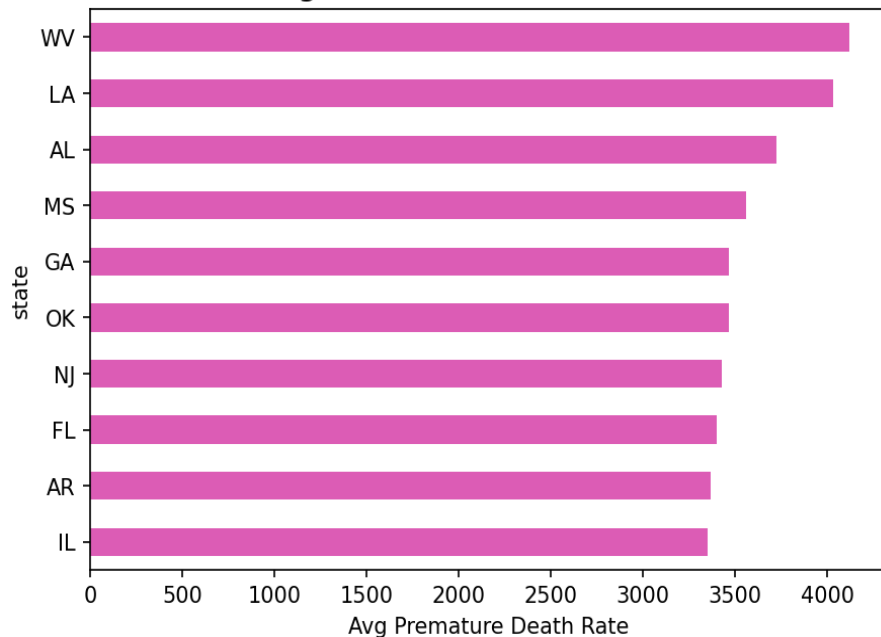
Premature Death Rate by State

10 Healthiest States (Lowest Premature Death Rate)



Healthiest: ID, HI, CO, WA, UT — lower inactivity, higher access to care

10 Least Healthy States (Highest Premature Death Rate)

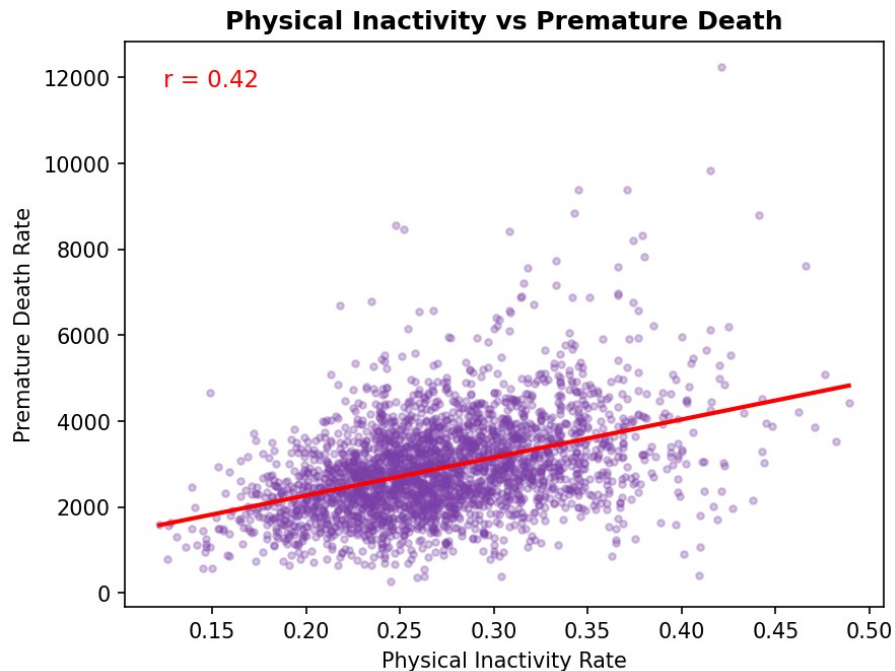


Least healthy: WV, LA, AL, MS, GA — higher poverty, inactivity, uninsured

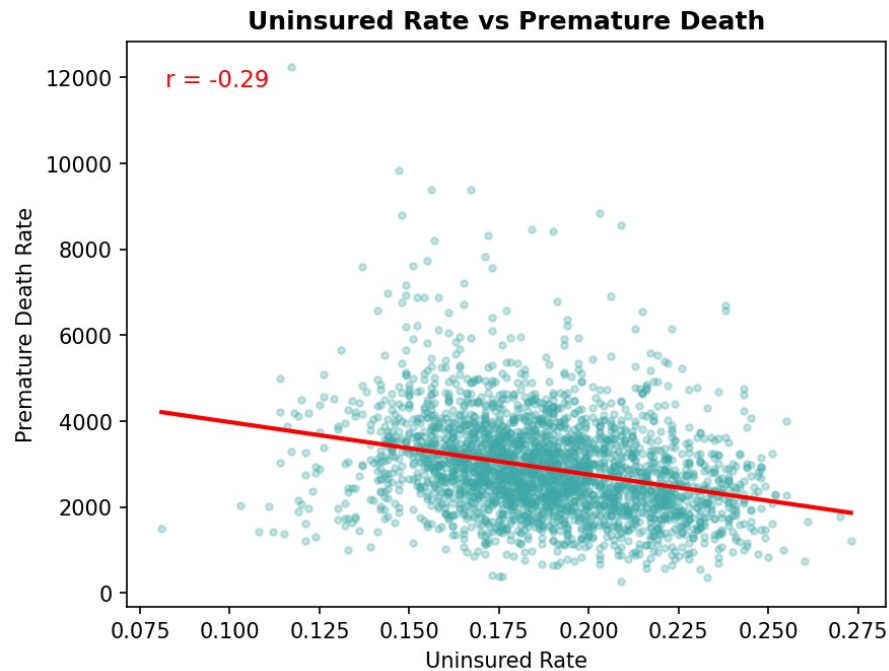
Results: Key Predictors

Physical inactivity and uninsured rates as predictors of premature death

Key Predictors of Premature Death Rate



Physical inactivity ($r=0.42$): Counties where more residents are physically inactive show consistently higher premature death rates.

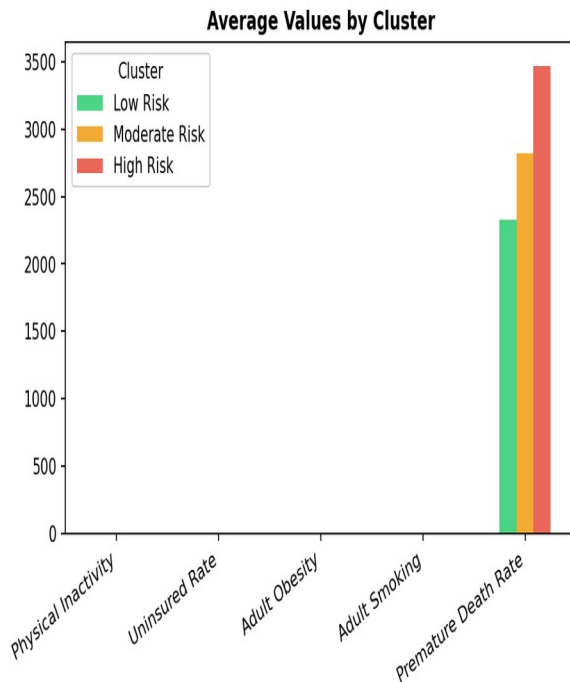
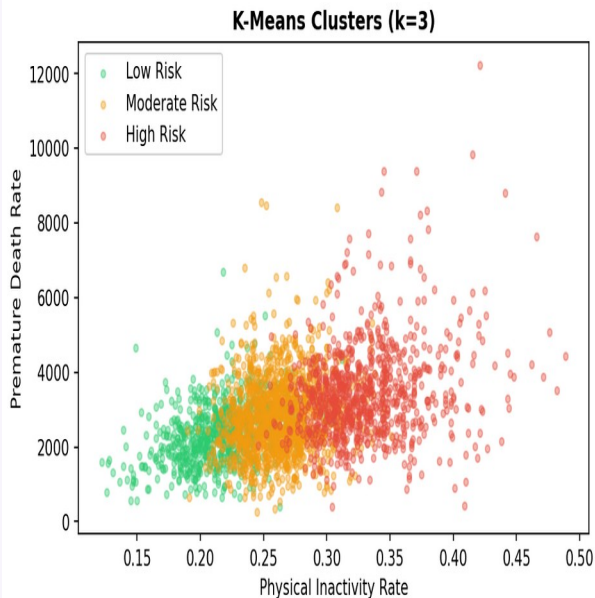


Uninsured rate ($r=-0.29$): Higher uninsured rates correlate with worse outcomes, reflecting reduced access to preventive care.

Results: K-Means Clustering (k=3)

Counties group into distinct health risk profiles

County Health Risk Clustering



Low Risk

640 counties | Avg death rate: 2,327

Lower inactivity, lower smoking, better access to care

Moderate Risk

1,344 counties | Avg death rate: 2,823

Average across all risk factors

High Risk

859 counties | Avg death rate: 3,468

Higher inactivity, obesity, smoking, uninsured rates

Features used for clustering: Physical Inactivity, Uninsured Rate, Adult Obesity, Adult Smoking

Challenges

Difficulties encountered during the project

FIPS Code Formatting

Each dataset used different county identifier formats — integers vs. zero-padded 5-character strings. Required standardization before merging.

Incomplete AQI Coverage

EPA air quality data only covers counties with active monitoring stations (~1,000 of 3,100+), leaving most counties without AQI values. Treated as optional feature.

CDC API Access Restrictions

The CDC PLACES API required pagination and rate-limiting logic to retrieve all county-level records in JSON format without errors.

Missing Data Handling

Some smaller/rural counties lacked values for several measures. Chose to drop rows with missing key variables rather than impute, reducing dataset from 3,152 to 2,843 counties.